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Editors-in-Chief
Computers & Geosciences

Dear Editors-in-Chief,

Please consider the manuscript entitled "A reproducible benchmark for transformation uncertainty in aquifer-test parameter inference" as an original research article for publication in *Computers & Geosciences*.

The manuscript addresses a computational hydrogeology problem at the interface between numerical groundwater modeling, analytical aquifer-test interpretation, and uncertainty quantification. Aquifer-test transmissivity and storage are commonly used as inputs to groundwater models and engineering calculations, but those quantities are inferred through interpretation models rather than observed directly. The paper provides a reproducible benchmark workflow for quantifying the transformation uncertainty introduced during this drawdown-to-parameter conversion.

The contribution combines a 10,000 scenario groundwater-flow benchmark, analytical limiting checks, grid-sensitivity checks, four analytical interpretation pathways, field diagnostics for two public pumping-test cases, and model-averaged hydraulic capacity and response-time diagnostics. The computational element is central to the paper: the benchmark supplies support-matched reference parameters and scenario-conditioned model factors, while the field workflow transfers those factors only where response diagnostics support comparable benchmark classes.

The submission follows the *Computers & Geosciences* checklist. It is single column and double spaced, uses author-date references, includes highlights, an authorship statement, and a code availability section, and provides a public GitHub repository together with a HydroShare archive for the larger generated data products.

The author confirms that this manuscript has not been published elsewhere and is not under consideration by another journal. The author approves the manuscript and agrees with its submission to *Computers & Geosciences*. No related manuscript is currently under consideration elsewhere.

Sincerely,

Ying-Fan Lin
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